



EKF for parameter estimation: 1a–1b

- You have learned how to use SPKF for parameter estimation.
- Now, you will learn how to use EKF for parameter estimation.

EKF step 1a: Parameter prediction time update.

- Due to the linearity of the parameter dynamics equation, we have $\hat{\theta}_k^- = \hat{\theta}_{k-1}^+$ (same as for SPKF).

EKF step 1b: Error covariance time update.

- Again, due to the linearity of the parameter dynamics equation, we have:

$$\Sigma_{\tilde{\theta},k}^- = \Sigma_{\tilde{\theta},k-1}^+ + \Sigma_{\tilde{r}}.$$

(also same as for SPKF).



EKF for parameter estimation: 1c

EKF step 1c: Output prediction.

- The system output is predicted to be:

$$\begin{aligned} \hat{d}_k &= \mathbb{E} \left[\begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix} \middle| \mathbb{D}_{k-1} \right] \\ &\approx \begin{bmatrix} f(x_{k-1}, u_{k-1}, \hat{\theta}_k^-, \bar{e}_k) \\ h(x_k, u_k, \hat{\theta}_k^-, \bar{e}_k) \end{bmatrix}. \end{aligned}$$

- That is, it is assumed that propagating $\hat{\theta}_k^-$ and the mean estimation error is the best approximation to predicting the output.



EKF for parameter estimation: 2a

EKF step 2a: Estimator gain matrix.

- The output prediction error may then be approximated:

$$\tilde{d}_k = d_k - \hat{d}_k = \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix} - \begin{bmatrix} f(x_{k-1}, u_{k-1}, \hat{\theta}_k^-, \bar{e}_k) \\ h(x_k, u_k, \hat{\theta}_k^-, \bar{e}_k) \end{bmatrix}$$

using a Taylor-series expansion on the first term:

$$\begin{aligned} d_k &\approx \begin{bmatrix} f(x_{k-1}, u_{k-1}, \hat{\theta}_k^-, \bar{e}_k) \\ h(x_k, u_k, \hat{\theta}_k^-, \bar{e}_k) \end{bmatrix} \\ &+ \underbrace{\frac{d}{d\theta} \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix} \bigg|_{\hat{\theta}_k^-}}_{\text{Defined as } \hat{C}_k^\theta} (\theta - \hat{\theta}_k^-) + \underbrace{\frac{d}{de_k} \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix} \bigg|_{\bar{e}_k}}_{\text{Defined as } \hat{D}_k^\theta} (e_k - \bar{e}_k). \end{aligned}$$



EKF for parameter estimation: 2a (cont.)

- From this, we can compute such necessary quantities as:

$$\begin{aligned}\Sigma_{\tilde{d},k} &\approx \hat{C}_k^\theta \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T + \hat{D}_k^\theta \Sigma_{\tilde{e}} (\hat{D}_k^\theta)^T \\ \Sigma_{\tilde{\theta}\tilde{d},k}^- &\approx \mathbb{E}[(\tilde{\theta}_k^-)(\hat{C}_k^\theta \tilde{\theta}_k^- + \hat{D}_k^\theta \tilde{e}_k)^T] \\ &= \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T.\end{aligned}$$

- These terms may be combined to get the Kalman gain:

$$L_k^\theta = \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T [\hat{C}_k^\theta \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T + \hat{D}_k^\theta \Sigma_{\tilde{e}} (\hat{D}_k^\theta)^T]^{-1}.$$



EKF for parameter estimation: 2b–2c

EKF step 2b: Parameter estimate measurement update.

- Compute *a posteriori* parameter estimate by updating *a priori* prediction using estimator gain and output innovation $d_k - \hat{d}_k$:

$$\hat{\theta}_k^+ = \hat{\theta}_k^- + L_k^\theta (d_k - \hat{d}_k).$$

EKF step 2c: Error covariance measurement update.

- Finally, the updated covariance is computed as:

$$\Sigma_{\tilde{\theta},k}^+ = \Sigma_{\tilde{\theta},k}^- - L_k^\theta \Sigma_{\tilde{d},k} (L_k^\theta)^T.$$



Summary

- You have now learned how to use EKF for parameter estimation when the state of the system is known.
- We initialize parameter estimate with best information re. the parameter value: $\hat{\theta}_0^+ = \mathbb{E}[\theta_0]$, and parameter estimation error covariance matrix:

$$\Sigma_{\tilde{\theta},0}^+ = \mathbb{E}[(\theta - \hat{\theta}_0^+)(\theta - \hat{\theta}_0^+)^T].$$

- A summary of the method is listed in the appendix.



Nonlinear EKF for parameter estimation (1)

State-space model:

$$\begin{aligned}\theta_{k+1} &= \theta_k + r_k, \\ d_k &= \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix}\end{aligned}$$

where r_k and e_k are independent Gaussian noise processes with means zero and $\bar{e}$, respectively, and having covariance matrices $\Sigma_{\tilde{r}}$ and $\Sigma_{\tilde{e}}$, respectively

Definitions:

$$\hat{C}_k^\theta = \frac{d \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix}}{d\theta} \bigg|_{\theta=\hat{\theta}_k^-} \quad \hat{D}_k^\theta = \frac{d \begin{bmatrix} f(x_{k-1}, u_{k-1}, \theta, e_k) \\ h(x_k, u_k, \theta, e_k) \end{bmatrix}}{de_k} \bigg|_{e_k=\bar{e}_k}$$



Nonlinear EKF for parameter estimation (2)

Initialization: For $k = 0$, set

$$\begin{aligned}\hat{\theta}_0^+ &= \mathbb{E}[\theta_0] \\ \Sigma_{\tilde{\theta},0}^+ &= \mathbb{E}[(\theta_0 - \hat{\theta}_0^+)(\theta_0 - \hat{\theta}_0^+)^T]\end{aligned}$$

Computation: For $k = 1, 2, \dots$ compute:

$$\begin{aligned}\text{State time update:} & \quad \hat{\theta}_k^- = \hat{\theta}_{k-1}^+ \\ \text{Covariance time update:} & \quad \Sigma_{\tilde{\theta},k}^- = \Sigma_{\tilde{\theta},k-1}^+ + \Sigma_{\tilde{r}} \\ \text{Output prediction} & \quad \hat{d}_k = \begin{bmatrix} f(x_{k-1}, u_{k-1}, \hat{\theta}_k^-, \bar{e}_k) \\ h(x_k, u_k, \hat{\theta}_k^-, \bar{e}_k) \end{bmatrix} \\ \text{Kalman gain matrix:} & \quad L_k^\theta = \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T [\hat{C}_k^\theta \Sigma_{\tilde{\theta},k}^- (\hat{C}_k^\theta)^T + \hat{D}_k^\theta \Sigma_{\tilde{e}} (\hat{D}_k^\theta)^T]^{-1} \\ \text{State measurement update:} & \quad \hat{\theta}_k^+ = \hat{\theta}_k^- + L_k^\theta (d_k - \hat{d}_k) \\ \text{Covariance meas. update:} & \quad \Sigma_{\tilde{\theta},k}^+ = \Sigma_{\tilde{\theta},k}^- - L_k^\theta \Sigma_{\tilde{d},k} (L_k^\theta)^T\end{aligned}$$